

From Attitudes to Behaviour: How Math Attitudes Influence Students' Behavioural Choices in Primary School

Overview

Research questions or hypotheses

To date, mathematics-related behaviors such as avoidance have primarily been studied through self-report measures or long-term outcomes like course selection and career choices (Pizzie & Kraemer, 2017). However, in primary school, where opting out of mathematics is not feasible, alternative approaches are needed to capture students' real-time behavioural choices (Hilz et al., 2025). The present study introduces an innovative behavioral measure by observing choices during a mathematical task, making it exploratory for its ordinal structure while hypothesizing monotonic task engagement from lower (skipping) to higher (self-reliance) levels. For further details about study design and method, see the section below. The aim of this study is to investigate the two following closely related research questions within an ordinal modelling framework, for which statistical power is planned. RQ1 examines how trait- and state-level math attitudes predict those choices and whether state-level attitudes mediate the effects of their trait-level counterparts, whereas RQ2 concerns the ordinal structure of behavioural choices. The two questions are therefore considered together, as they address the same behavioural outcome from complementary perspectives. RQ1: Whether the following math attitudes (avoidance, buoyancy, worry and perceived competence) predict students' behavioural choices during the mathematical task (i.e., skipping, help-seeking, and self-reliant responding), controlling for gender and math ability, and do state-level attitudes statistically account for part of the association between their corresponding trait-level measures and these choices? HP1: Given the multiple data-dependent decisions involved, the analyses addressing RQ1 are best characterised as inferentially exploratory rather than strictly confirmatory. Specifically, directional hypotheses are preregistered a priori, but their formal testing depends on a sequence of model-selection and predictor-screening decisions that cannot be fully fixed in advance. We therefore treat the final p-values as conditional on the selected model and acknowledge that this procedure provides only imperfect control over false positive results. Specifically, while the direction of expected effects is grounded in converging evidence (Choe et al., 2019; Putwain et al., 2020; Wang et al., 2025), the final set of predictors, the mediation structure, and the response parameterisation cannot be fully determined in advance. Predictor inclusion will follow the model-building procedure described in the "Statistical models" section. Math anxiety and self-competence are hypothesised to influence behavioural choices primarily through their state-level counterparts (math anxiety via worry, and self-competence via perceived competence), whereas buoyancy and avoidance, if psychometrically adequate, are expected to contribute directly to choice. After accounting for prior math ability and gender, students with more adaptive motivational-affective profiles are expected to show greater task engagement. Specifically, higher academic buoyancy and higher perceived competence are expected to predict more engaged behavioural choices (i.e., self-reliant responding), whereas higher avoidance and higher worry are expected to predict less engaged choices (i.e., skipping). Prior math ability from T0 and gender will be included as covariates in all models to distinguish ability-driven from attitude-driven variation in choices. The two-stage model-building procedure described in the Statistical models section is designed to detect and accommodate such patterns. RQ2: Within the ordinal representation of students' behavioural choices during the mathematical task (skipping < help-seeking < self-reliance), do some predictors require

category-specific effects across ordinal transitions? HP2: We expect students' behavioural choices to be compatible with an ordered representation of task engagement, in which skipping reflects lower engagement, help-seeking an intermediate form, and self-reliant responding higher engagement. These effects are expected to be monotonic across the ordinal continuum from skipping to help-seeking to self-reliance. However, we acknowledge that the proportional-odds assumption may not hold uniformly for all predictors: some attitude variables may differentiate more strongly between specific transitions than others, warranting category-specific parameterisation for those predictors. If the proportional-odds assumption is not supported for one or more predictors, category-specific ordinal parameterisations will be estimated. Inferential tests will be conducted only after model selection has been completed according to the preregistered criteria. All inferential results will therefore be explicitly reported as conditional on the selected model and should not be interpreted as if the final model had been fixed a priori.

Foreknowledge of data or evidence

Data exists but the authors have not observed it yet. At least some of the data that will be used for this analysis plan exists and is possible for the authors to access. However, the authors certify that they have not accessed any of that data and will not do so until after this plan is registered.

Explanation of foreknowledge and managing unintended influences

Data collection is ongoing and is expected to conclude in late May 2026. This registration was completed prior to any human observation of the raw data. Registration is prior to any human observation of the raw data.

Research Design

Study type

Non-randomized study: Does not include random assignment of subjects to treatments or conditions. This usually includes correlational studies, observational studies, surveys, natural experiments, real-world evidence studies, regression discontinuity designs, and descriptive studies.

Intention for causal interpretation

Indirect inference on causal relationship(s): This study is intended to examine a potential causal relationship, but is not designed to isolate the specific causal effect. This work may inform causal inference indirectly or advance insight toward future estimation of a causal relationship.

Blinding of experimental treatments

No blinding is involved.

Additional blinding during research or analysis

No data

Study design

The study employs two sessions (T0 and T1) to examine the relationships between math attitudes (both trait- and state-level), math behavioural choice, and math ability. At T0,

participants complete a first block of 16 arithmetic problems, followed by two state questionnaires assessing worry and perceived competence. The session then continues with a second block of 16 arithmetic problems. At the end of the session, participants complete two trait questionnaires assessing math anxiety and academic buoyancy. At T1, participants complete 32 arithmetic problems divided into two equal blocks. Unlike T0, each problem is presented together with three response options corresponding to different behavioural choices: participants may skip the item and move on to the next one, request a hint before attempting the problem, or solve the problem independently. After completing both blocks of the math task, participants complete the remaining two trait questionnaires assessing math avoidance and math self-competence. All tasks are administered by computer in small groups within the classroom. For both RQ1 and RQ2, the primary outcome variable is students' item-level behavioural choice during the math task (i.e., skipping, help-seeking, or self-reliance behaviour). The primary working assumption is that behavioural choice can be represented as an ordinal outcome (skipping < help-seeking < self-reliance). RQ2 evaluates whether this ordinal representation is adequate under proportional odds or whether selected predictors require category-specific effects. Trait- and state-level math attitudes will be included as predictors. No special actions will be taken by the experimenter for participants showing extreme patterns (e.g., skipping all items). Such behaviour is discouraged by two task design features: (1) a mandatory 15-second dwell time per item after each choice, and (2) automated prompts every 5 consecutive skips or help requests, reminding participants of all the available behavioural choices. If the hypothesised ordinal structure proves clearly inadequate, additional follow-up analyses may be conducted to explore whether a categorical specification provides a more appropriate representation of the data. These analyses are not part of the preregistered decision pipeline, are not implemented in the current analysis script, and will be reported explicitly as supplementary, non-preregistered analyses.

No files selected

Randomization

No data

Sampling

Data collection procedures

Participants are recruited by initially contacting the Heads of Public Primary Schools in middle-income areas of a Western European country. Once the teachers grant permission, parental consent is requested. The study involves students in the 4th and 5th grades, aged 9 to 11 years. Exclusion criteria include formal diagnoses of intellectual disabilities or neurological/genetic conditions. No financial compensation will be provided. The study has received approval from the Authors' University ethics committee for psychological research. Data collection takes place during the second half of academic year (2025-2026). The first session (T0) is conducted starting in early March 2026, followed by the second session (T1), which will be conducted between mid-April and late May 2026.

Data collection procedures - File upload

No files selected

Sample size

The sample size is partially constrained by available resources. We aim to recruit approximately 35-40 classes, corresponding to an initial target of about 500 students. We anticipate a missing rate of approximately 8% per session, due primarily to school absences (on average, 1-2 out of 20 children per class on a given testing day). Additionally, we estimate that approximately 15-20 children per class will meet the inclusion criteria and return the signed parental informed consent. Based on these factors, we expect to include 400 eligible participants in the final analysis. Recruitment will continue until 400 eligible participants are obtained or all recruited classes complete both sessions, whichever occurs first. If resources limit recruitment below 400, analysis proceeds with the achieved $N \geq 300$, where simulations confirm $> 90\%$ power (ordinal conceptualization) or adequate Monte Carlo-based standard-error precision for the primary parameters (categorical conceptualization) at $N=300$.

Sample size rationale

Two simulation-based sample-size analyses were conducted: one for the primary ordinal analysis and one for a possible multinomial robustness analysis in case the ordinal representation proves inadequate. Full details and the power analysis code are provided in the attached files “PowerAnalysisOrdinal.R” (ordinal conceptualization) and “FastPowerCategorical.R” (categorical conceptualization). In both cases, the power analyses are fully aligned with the planned statistical models outlined in the “Analysis Plan section” and are based on data-generating processes consistent with the large simulated dataset attached in “Study Design section”. The model run in both power analyses includes all the possible predictors and the covariates to be as conservative as possible. The following paragraphs provide a concise overview of both power analysis procedures and their results. 1) Power analysis for the ordinal conceptualization of the dependent variable (math-related behavioural choices) was evaluated using a simulation-based approach with cumulative link mixed models (CLMM). The power analysis implemented in “PowerAnalysisOrdinal.R”, was based on a power simulation with 5,000 iterations Monte Carlo simulation iterations of the full model, including predictors relevant to both RQ1 and RQ2. The simulated logistic linear regression model tested the effects of standardized predictors (buoyancy, avoidance, worry, perceived competence) on ordered behavioural choices collected at T1 (skipping < help-seeking < self-reliance), while controlling for all relevant covariates and random intercept, thus accounting for intersubjective variability. Under the alternative hypothesis, based on coefficients typically reported in literature, betas were set to -0.25 for worry, -0.30 for avoidance, 0.30 buoyancy, and 0.25 for perceived competence, and a subject-level random intercept variance of $\tau = 1$, reflecting moderate individual differences between subjects in baseline behavioural choice disposition. These values correspond to statistically small effects, consistent with those typically observed in psychological research (odds ratios ≈ 0.78 for worry/avoidance, ≈ 1.35 for buoyancy/self-competence per 1 SD), equivalent to an effect size of Cohen's $d \approx 0.17$ using standard logistic conversion formulas. The simulation incorporated the planned multilevel structure of the study: students aged 9-11 years (target $N=400$), items nested within students (64 items total, 32 per session with T1 as outcome), and realistic trait-state correlations via a multivariate covariance matrix. Results for H1 indicated power exceeding 80% for detecting each individual predictor effect at $\alpha = 0.05$ (directional, one-sided tests) at $N=300$. Joint power for detecting all four predictors significant simultaneously was approximately 83% without correction, remaining at $\sim 83\%$ after false discovery rate (FDR) correction across the four predictors (buoyancy, avoidance, worry, perceived competence). In order to prove that the simulated process is valid and well-calibrated, null hypothesis (H_0) simulations have been conducted with all betas = 0, confirming Type I error rates near 5%. This result demonstrated no true effects present under

the null hypothesis scenario, confirming no inflation of false positives. 2) Power analysis for the categorical conceptualization was conducted using simulation-based methods with multilevel multinomial logistic regression (brms). Due to excessive computational demands for the full model, a streamlined power analysis was implemented. The power analysis implemented in “FastPowerCategorical.R” simulates data across a grid of sample sizes (N=100 to 500 in steps of 50), averaging 10 replications per N to compute variance-covariance matrices of key posterior parameters for RQ2. The simulated model tested the effects of standardized predictors (buoyancy, avoidance, worry, and perceived competence) on categorical behavioural choices at T1 (skip, help-seeking, self-reliance), while controlling for prior math ability at T0, math anxiety, self-competence, gender, and random intercepts. Under the alternative hypothesis (H1), logit coefficients for key contrasts were set to reflect plausible shifts in choice probabilities. For skip vs. (help-seeking/self-reliance): buoyancy $\beta = -0.30$, avoidance $\beta = +0.30$, worry $\beta = +0.25$, perceived competence $\beta = -0.25$. For help-seeking vs. self-reliance: smaller effects in all the considered predictors (e.g., buoyancy $\beta = -0.15$). These yield odds ratios which will range ≈ 0.74 -1.35 per 1 SD across primary contrasts, equivalent to an effect size of Cohen's $d \approx 0.17$ using standard logistic conversion formulas. As before, the simulation incorporated the planned multilevel structure of the study, with items nested within participants (32 items per participant at session 2 as outcome, across approximately 400 participants aged 9-11 years), and realistic trait-state correlations via a multivariate covariance matrix. Power analysis prioritized the four primary contrasts central to HP2 (effects of buoyancy, avoidance, worry, and perceived competence on self-reliance vs. skipping behaviour). Secondary help-seeking parameters were monitored for completeness, focusing especially on one additional parameter (buoyancy effect on help-seeking). To determine a plausible sample size, we first identified target standard errors for the primary effects after Bonferroni correction across five predictors, corresponding to 80% power. Across a simulation grid (N=100-500), posterior SEs for the four primary parameters fell below target thresholds (ranging from 0.035-0.079) by N=300: effects of buoyancy (0.033), avoidance (0.016), perceived competence (0.032), and worry (0.067). The secondary parameter (buoyancy effect on help-seeking) remained slightly elevated (0.068) compared to its stringent target (0.035) due to its smaller effect simulated ($\beta = 0.15$) and the conservative Bonferroni correction across five predictors, which is known to reduce power for smaller effects. In this context, MCMC is used to obtain stable coefficient and standard-error estimates for complex multinomial mixed models. The resulting quantities are used for approximate frequentist evaluation of precision and test performance, not for Bayesian posterior inference. Further details on these assumptions and the complete R code for the power analysis can be found in the attached files “PowerAnalysisOrdinal.R” and “FastPowerCategorical.R”.

Starting and stopping rules

Data collection will stop when either (a) 400 eligible participants with data from both sessions have been obtained, or (b) all previously recruited classes have completed both sessions, whichever occurs first. No interim analyses of the study hypotheses will be conducted before the end of data collection.

Variables

Manipulated variables

None

No files selected

Measured variables

The experiment includes three main sets of measured variables: math performance, math behavioural choices, trait- and state-level math attitudes. Math Performance: is assessed in both sessions (T0 and T1) using two parallel digital algebraic tasks each consisting of 32 different problems (e.g., $34 - ? = 21$). A pool of 100 arithmetic problems has been generated using a custom R script, subject to several constraints (e.g., operands ranging from 12 to 99, exclusion of multiples of 10, and balanced positioning of the unknown term across items). The generated pool has been subsequently reviewed and fine-adjusted. For each participant, 32 problems per session are randomly sampled from this pool. To prevent repetitions, the sampling algorithm excludes problems already presented in T0 when drawing items for T1. For each item, students have a time limit of 25 seconds, determined through a prior pilot test, to solve the problems and enter the correct answer using the keyboard. 1 point will be assigned for correct and timely answers, while 0 points for incorrect, late or skipped responses. Math behavioural choices: will be assessed only in the second session (T1). For each item, participants will select one of three task-related options: (1) not solving the problem, indicating skipping behaviour; (2) requesting a hint, that provides a brief problem-solving strategy (e.g., for “ $? - 34 = 16$ ”, the hint would be “Try using the strategy: $34 + 16 = ?$ ”), indicating help-seeking behaviour; or (3) solving the problem independently, reflecting self-reliance behaviour. For each item, the selected behavioural choice yields one categorical choice variable with three levels (skip, help or self), yielding an objective measure of students’ observable behavioural choices adapted from the paradigm proposed by Wang et al. (2025). Trait measures will be assessed using the following self-report questionnaires: - Math anxiety: will be measured using the Abbreviated Mathematics Anxiety Scale for Children (AMAS, Caviola et al., 2017). The 9-item measure assesses participants’ anxiety levels towards mathematics in different scenarios on a 5-point Likert scale, with higher scores indicating greater math anxiety. - Academic Buoyancy: will be measured using the Academic Buoyancy Scale (Martin & Marsh, 2008). The 4-item questionnaire measures students’ ability to cope with everyday academic challenges using a 7-point Likert scale, with higher scores indicating greater academic buoyancy. - Math avoidance: will be assessed using an adapted version of the Math Avoidance Scale (Conlon et al., 2022). The questionnaire measure participants’ avoidant thoughts or behaviours in 10 math-related situations on a 4-point Likert scale, with higher scores indicating greater math avoidance. - Math Self-competence: will be assessed using an adapted version of Harter’s questionnaire (2012), measuring scholastic self-competence in the mathematical domain. The final 10-item version measure uses a 4-point Likert scale, with higher scores indicating greater math self-competence. An additional variable is also collected at T0, namely trait test anxiety; however, this is not included or analysed in the present study and will be used for other purposes in a different study. State-level math attitudes will be assessed using the following self-report questionnaires administered in close temporal proximity to math performance at T0. - Worry: will be assessed using three adapted items from Mammarella et al. (2023). Items measure participants’ worry levels during the math task on a 4-point Likert scale, with higher scores indicating greater worry. - Perceived Competence: will be assessed using three adapted items from Mammarella et al. (2023). Items measure participants’ perceived competence during the math task on a 4-point Likert scale, with higher scores indicating greater perceived competence.

Measured variables - File upload

[Data dictionary.csv](#)

Indices

For screening measures, a total math score will be calculated based on accuracy in the math task at T0. Sum scores will be used for all trait and state questionnaires. Math behavioural choices will be coded at the item level as described in the “Measured variables” section.

Indices - File upload

No files selected

Analysis Plan

Statistical models

This section provides a brief overview of the data analysis steps and modelling approach. The full analysis pipeline, including data preparation, model specifications, and hypothesis tests, is implemented in the attached R script “PrePlannedAnalysis.R”, which is applied to a simulated dataset (“SimulatedDatasetBehaviours.csv”). Statistical modelling and hypothesis testing Prior to structural modelling, a confirmatory factor analysis (CFA) will be conducted to evaluate the psychometric adequacy of buoyancy and avoidance as latent constructs. Initial model fit will be evaluated against the pre-specified criteria (CFI > .95; RMSEA < .08). If the initial model does not meet these criteria, modification indices may be inspected in a limited and theory-constrained way. At most two correlated residuals will be added, and only when they involve item pairs with clear substantive or wording overlap. Any such re-specification will be documented explicitly and treated as a pragmatic measurement adjustment rather than as confirmatory evidence for the revised measurement model. Reliability will be evaluated using ordinal alpha and treated as a descriptive indicator of measurement adequacy rather than as a strict pass/fail criterion. Values clearly below conventional standards will be discussed as a limitation and may motivate sensitivity analyses excluding the corresponding construct. Additionally, prior to including both constructs in the structural model, multicollinearity between buoyancy and avoidance will be assessed; if the inter-construct correlation exceeds $|r| > .70$, the two predictors will not be entered simultaneously in the same primary model. In that case, separate sensitivity models will be estimated for each construct, and the main confirmatory inference will be restricted to predictors that can be included without problematic redundancy. If one or both constructs fail the pre-specified fit criteria, the corresponding variable(s) will be excluded from all structural models, and this exclusion will be transparently reported. In this case, the primary analysis will proceed with the retained predictors only. RQ1. The model-building strategy then proceeds in three sequential steps. First, within the ordinal family, the predictor structure will be established through nested cumulative logit mixed-effects models estimated in brms. The baseline model (M0) includes only state-level predictors (worry and perceived competence), the covariates (prior math ability at T0 and gender), and a random intercept for participant. This constitutes the full mediation model. M1 extends M0 by adding buoyancy and avoidance as direct predictors of choice, conditional on satisfactory psychometric screening and acceptable multicollinearity. M2 extends M1 by adding direct paths from math anxiety and self-competence to choice, thereby testing partial against full mediation. M2 will be estimated only if M1 is preferred over M0. RQ2. Starting from the Stage-1 selected ordinal model, category-specific extensions will be evaluated by relaxing the proportional-odds constraint for selected predictors and comparing each extension against its predecessor via LOO. In the current analysis script, this step is implemented as a worked example and template for the same comparison logic. Each extension will be compared against its predecessor via LOO, and a category-specific

parameterisation will be retained only if the more complex model is clearly preferred. The model retained at the end of the preregistered selection pipeline will be used for the inferential tests, and all inferential results will be reported as conditional on that selected model. For the selected best-fitting model, directional inferences will be conducted for each retained primary predictor. For predictors satisfying proportional odds, a single coefficient will be interpreted. For predictors with category-specific effects, coefficients will be interpreted separately. Because the analysis combines preregistered model selection with MCMC-based estimation, all inferential results will be interpreted within a pragmatic frequentist framework and treated as conditional on the selected model. MCMC is used here as a computational device to approximate parameter estimates and standard errors in models for which simpler estimation routines are not readily available, not as a basis for Bayesian inference. Multiplicity adjustment will be applied only across the retained primary directional effects using the FDR procedure. Secondary and exploratory parameters will be reported descriptively and clearly distinguished from the preregistered primary effects.

Statistical models - File upload

[PrePlannedAnalysis.R](#)[PowerCategorical.zip](#)[PowerOrdinal.zip](#)[SimulatedDatasetBehaviours.csv](#)

Transformations

Data will be imported and pre-processed to include trials from T1, which constitute the outcome of interest. Measures of math performance and math attitudes assessed both at T0, including both trait- and state-level variables, as well as additional trait-level measures collected at Session 2 (T1), will be merged with the behavioural data and included as predictors or covariates. For each item at T1, a single outcome variable (choice) will be derived, indicating the behavioural option selected by the student (skipping, help-seeking, self-reliance). All continuous predictors will be standardized (z-scores).

Inference criteria

To address both RQs, the model with higher ELPD under Pareto-smoothed LOO-CV will be selected as best-fitting (see "PrePlannedAnalysis.R"). Specifically, the differential ELPD must exceed twice its standard error, consistent with conventional LOO-CV thresholds (Vehtari et al., 2017). This applies to all pairwise model comparisons in the two-stage model-building procedure: M0 vs. M1, M1 vs. M2 (Stage 1), and each successive extension vs. its predecessor (Stage 2). At each step, if the more complex model is not clearly preferred, the simpler model is retained, and the pipeline does not proceed to the next extension. To address RQ1, effects will be evaluated using one-tailed p-values based on MCMC-estimated coefficients and their associated standard errors. For the primary directional hypotheses, p-values < .05 after multiplicity correction will be treated as statistically significant. These p-values are interpreted as approximate frequentist quantities, conditional on the selected model, and should be read with caution given the data-dependent model-selection procedure. Accordingly, the inferential claims are modest: statistical significance is used as one decision criterion, not as definitive evidence, and residual inflation of false positives cannot be ruled out. Directional expectations are specified a priori: positive coefficients expected for academic buoyancy and perceived competence (reflecting a higher probability of self-reliant behaviour relative to help-seeking and skipping), and negative for avoidance and worry (reflecting a higher probability of skipping relative to help-seeking and self-reliant behaviour). If one or both constructs fail the pre-

specified fit criteria, the corresponding variable(s) will be excluded from all structural models, and this exclusion will be transparently reported. In this case, the primary analysis will proceed with the retained predictors only. For the selected best-fitting model, directional hypothesis tests will be conducted for each primary predictor retained in the final model. The number of primary comparisons ranges from two to four, depending on the inclusion of buoyancy and avoidance (conditional on measurement model fit). Since RQ1 tests, potentially, all four predictors simultaneously, p-values for the preregistered primary directional effects retained in the final selected model will be adjusted using the False Discovery Rate (FDR) procedure. We note, however, that because predictor inclusion and model structure are themselves data-dependent, this correction only partially addresses the overall false positive risk and does not fully account for the full analytic search process. Indirect effects will be estimated by multiplying posterior draws from the relevant trait-to-state model (path a) by posterior draws from the corresponding coefficient(s) in the selected behavioural-choice model (path b). These indirect effects will be interpreted cautiously as conditional statistical decompositions rather than as strong causal mediation effects, given the partially time-lagged and partially concurrent measurement structure of the study.

Data inclusion and exclusion

Participants will be excluded from the primary analysis if they are missing the entire T1 behavioural task, if they lack the T0 math-ability covariate, or if technical problems prevent reconstruction of their item-level choices. Participants showing extreme but technically valid behavioural patterns (e.g., skipping all items) will not be excluded on that basis alone, because such patterns are substantively relevant to the research question.

Missing data

We expect a low rate of missing data for the questionnaire responses, as participants are forced to answer every item, with the task being administered on a computer. However, math trials in session 2 (32 items) may exhibit higher apparent missingness due to the explicit skip option alongside help-seeking and self-reliant choices. Skipped trials will not be treated as missing but will be retained as distinct behavioural category ("choice"=skip) in the outcome variable, with task accuracy coded as 0.

Other planned analysis

To inform future research, exploratory analyses may examine whether demographic characteristics are associated with the behavioural outcomes of interest or moderate the observed effects. These analyses are intended to be descriptive and hypothesis-generating and will be clearly distinguished from the preregistered confirmatory analyses.

Other

Context and additional information

No data